East West University

Department of Computer Science and Engineering

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Lab Assignment Submission Report-02

ASSIGNMENT TITLE: Intermediate level DDL, and Simple DML statements

Course Code: CSE302

Course Title: Database Systems

Section: 02

SUBMITTED BY

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DATE OF SUBMISSION

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Screenshots

Lab Task # 01 (Schema Definition)

i) account

```
SQL> CREATE TABLE account(
  2       account_no char(5),
  3       balance number NOT NULL,
  4       CONSTRAINT account_ACC PRIMARY KEY(account_no),
  5       CONSTRAINT account_BAC CHECK (balance>=0)
  6  );

Table created.
```

Figure: 01

ii) customer

```
SQL> CREATE TABLE customer(
  2       customer_no char(5),
  3       customer_name VARCHAR2(20) NOT NULL,
  4       customer_city VARCHAR2(10),
  5       CONSTRAINT customer_PK PRIMARY KEY(customer_no)
  6  );

Table created.
```

Figure: 02

iii) depositor

```
SQL> CREATE TABLE depositor(
  2      account_no char(5),
  3      customer_no char(5),
  4      CONSTRAINT PK_depositor PRIMARY KEY (account_no,customer_no)
  5  );

Table created.
```

Figure: 03

Lab Task # 02 (Schema Modification)

i) Write SQL statement to add a new attribute 'date_of_birth' (date type) in customer table.

```
SQL> ALTER TABLE customer ADD date_of_birth date;

Table altered.

SQL> desc customer;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 CUSTOMER_NO                               NOT NULL CHAR(5)
 CUSTOMER_NAME                             NOT NULL VARCHAR2(20)
 CUSTOMER_CITY                                      VARCHAR2(10)
 DATE_OF_BIRTH                                      DATE
```

Figure: 04

ii) Write SQL statement to drop the attribute ‘date_of_birth’ from customer table.

```
SQL> ALTER TABLE customer DROP COLUMN date_of_birth;

Table altered.

SQL> desc customer;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 CUSTOMER_NO                               NOT NULL CHAR(5)
 CUSTOMER_NAME                             NOT NULL VARCHAR2(20)
 CUSTOMER_CITY                                      VARCHAR2(10)
```

Figure: 05

iii) Write SQL statement to rename the attribute account_no, customer_no from depositor table to a_no and c_no, respectively.

```
SQL> ALTER TABLE depositor RENAME COLUMN account_no TO a_no;

Table altered.

SQL> ALTER TABLE depositor RENAME COLUMN customer_no TO c_no;

Table altered.

SQL> desc depositor;
 Name                                      Null?    Type
 ----------------------------------------- -------- ----------------------------
 A_NO                                      NOT NULL CHAR(5)
 C_NO                                      NOT NULL CHAR(5)
```

Figure: 06

iv) Write SQL statements to add two foreign key constraints ‘depositor_fk1’ and ‘depositor_fk2’ which identifies a_no and c_no as a foreign key.

```
SQL> ALTER TABLE depositor ADD CONSTRAINT  depositor_fk1 FOREIGN KEY(a_no) REFERENCES account;

Table altered.

SQL> ALTER TABLE depositor ADD CONSTRAINT  depositor_fk2 FOREIGN KEY(c_no) REFERENCES customer;

Table altered.


SQL> SELECT *FROM user_cons_columns
  2  WHERE TABLE_NAME='DEPOSITOR';

OWNER                          CONSTRAINT_NAME
------------------------------ ------------------------------
TABLE_NAME
------------------------------
COLUMN_NAME
--------------------------------------------------------------------------------
  POSITION
----------
JANNATUL                       PK_DEPOSITOR
DEPOSITOR
A_NO
         1


OWNER                          CONSTRAINT_NAME
------------------------------ ------------------------------
TABLE_NAME
------------------------------
COLUMN_NAME
--------------------------------------------------------------------------------
  POSITION
----------
JANNATUL                       PK_DEPOSITOR
DEPOSITOR
C_NO
         2
```

Figure: 07

```
OWNER                          CONSTRAINT_NAME
------------------------------ ------------------------------
TABLE_NAME
------------------------------
COLUMN_NAME
--------------------------------------------------------------------------------
  POSITION
----------
JANNATUL                       DEPOSITOR_FK1
DEPOSITOR
A_NO
         1


OWNER                          CONSTRAINT_NAME
------------------------------ ------------------------------
TABLE_NAME
------------------------------
COLUMN_NAME
--------------------------------------------------------------------------------
  POSITION
----------
JANNATUL                       DEPOSITOR_FK2
DEPOSITOR
C_NO
         1
```

Figure: 08

Lab Task # 03 (Inserting Records into Tables)

i) Account

```
SQL> INSERT INTO account VALUES('A-101', 12000);

1 row created.

SQL>
SQL> INSERT INTO account VALUES('A-102', 6000);

1 row created.

SQL>
SQL> INSERT INTO account VALUES('A-103', 2500);

1 row created.

SQL> SELECT * FROM account;

ACCOU    BALANCE
----- ----------
A-101      12000
A-102       6000
A-103       2500
```

Figure: 09

ii) Customer

```
SQL> INSERT INTO customer VALUES('C-101', 'Alice', 'Dhaka');

1 row created.

SQL>
SQL> INSERT INTO customer VALUES('C-102', 'Annie', 'Dhaka');

1 row created.

SQL>
SQL> INSERT INTO customer VALUES('C-103', 'Bob', 'Chittagong');

1 row created.

SQL>
SQL> INSERT INTO customer VALUES('C-104', 'Charlie', 'Khulna');

1 row created.

SQL> SELECT * FROM customer;

CUSTO CUSTOMER_NAME        CUSTOMER_C
----- -------------------- ----------
C-101 Alice                Dhaka
C-102 Annie                Dhaka
C-103 Bob                  Chittagong
C-104 Charlie              Khulna
```

Figure: 10

iii) Depositor

```
SQL> INSERT INTO depositor VALUES('A-101', 'C-101');

1 row created.

SQL>
SQL> INSERT INTO depositor VALUES('A-103', 'C-102');

1 row created.

SQL>
SQL> INSERT INTO depositor VALUES('A-103', 'C-104');

1 row created.

SQL>
SQL> INSERT INTO depositor VALUES('A-102', 'C-103');

1 row created.

SQL> SELECT * FROM depositor;

A_NO  C_NO
----- -----
A-101 C-101
A-102 C-103
A-103 C-102
A-103 C-104
```

Figure: 11

Lab Task # 04 (Queries)

i) Display customer name and customer city only.

```
SQL> SELECT customer_name,customer_city FROM customer;

CUSTOMER_NAME        CUSTOMER_C
-------------------- ----------
Alice                Dhaka
Annie                Dhaka
Bob                  Chittagong
Charlie              Khulna
```

Figure: 12

ii) Display the unique customer city. No repetitions are allowed.

```
SQL> SELECT distinct customer_city FROM customer;

CUSTOMER_C
----------
Dhaka
Khulna
Chittagong
```

Figure: 13

iii) Find account numbers with balance more than 7000.

```
SQL> SELECT account_no FROM account
  2  WHERE balance>7000;

ACCOU
-----
A-101
```

Figure: 14

iv) Find customer number and customer name who live in Khulna.

```
SQL> SELECT customer_no,customer_name FROM customer
  2  WHERE customer_city='Khulna';

CUSTO CUSTOMER_NAME
----- --------------------
C-104 Charlie
```

Figure: 15

v) Find customer number and customer name who do not live in Dhaka.

```
SQL> SELECT customer_no,customer_name FROM customer
  2  WHERE customer_city!='Dhaka';

CUSTO CUSTOMER_NAME
----- --------------------
C-103 Bob
C-104 Charlie
```

Figure: 16

vi) Find customer name and customer city who have accounts with balance more than 7000.

```
SQL> SELECT customer_name, customer_city
  2  FROM account acc, customer cst, depositor depst
  3  WHERE (acc.account_no = depst.a_no) AND (depst.c_no=cst.customer_no) AND balance>7000;

CUSTOMER_NAME        CUSTOMER_C
-------------------- ----------
Alice                Dhaka
```

Figure:17

vii) Find customer name and customer city who have accounts with balance more than 7000 and do not live in Khulna.

```
SQL> SELECT customer_name, customer_city
  2  FROM account acc, customer cst, depositor depst
  3  WHERE (acc.account_no = depst.a_no) AND (depst.c_no=cst.customer_no) AND balance>7000 AND (customer_city!='Khulna');

CUSTOMER_NAME        CUSTOMER_C
-------------------- ----------
Alice                Dhaka
```

Figure: 18

viii) Find account number and balance for those accounts which belong to a customer with id ‘C-102’.

```
SQL> SELECT account_no, balance
  2  FROM account acc, customer cst, depositor depst
  3  WHERE (acc.account_no = depst.a_no) AND (cst.customer_no = depst.c_no) AND c_no = 'C-102';

ACCOU    BALANCE
----- ----------
A-103       2500
```

Figure: 19

ix) Find all account number and balance for those accounts which belong to customers of Dhaka and Khulna city.

```
SQL> SELECT account_no, balance
  2  FROM account acc, customer cst, depositor depst
  3  WHERE (cst.customer_no=depst.c_no) AND (depst.a_no =a_no) AND (depst.a_no = acc.account_no)
  4  and (customer_city='Dhaka' or customer_city='Khulna');

ACCOU    BALANCE
----- ----------
A-101      12000
A-103       2500
A-103       2500
```

Figure: 20

x) Find the customer who have no accounts.

```
SQL> SELECT customer_name
  2  FROM account acc, customer cst, depositor depst
  3  where (cst.customer_no=depst.c_no) AND (depst.a_no = acc.account_no) AND acc.account_no= NULL;

no rows selected
```

Figure: 21

Learning Outcomes: The following is what we have learned in this lab.

1) We have learned to Schema Definition along with Integrity Constraints-Primary key, Foreign key, NOT NULL, Check and Unique constraints. And also create tables with these constraints.

2) We have learned Schema Modification- Adding, Dropping, Renaming and Deleting etc. Also learned to inserting records into tables and some queries.

3) Then finally, we have completed the lab work.